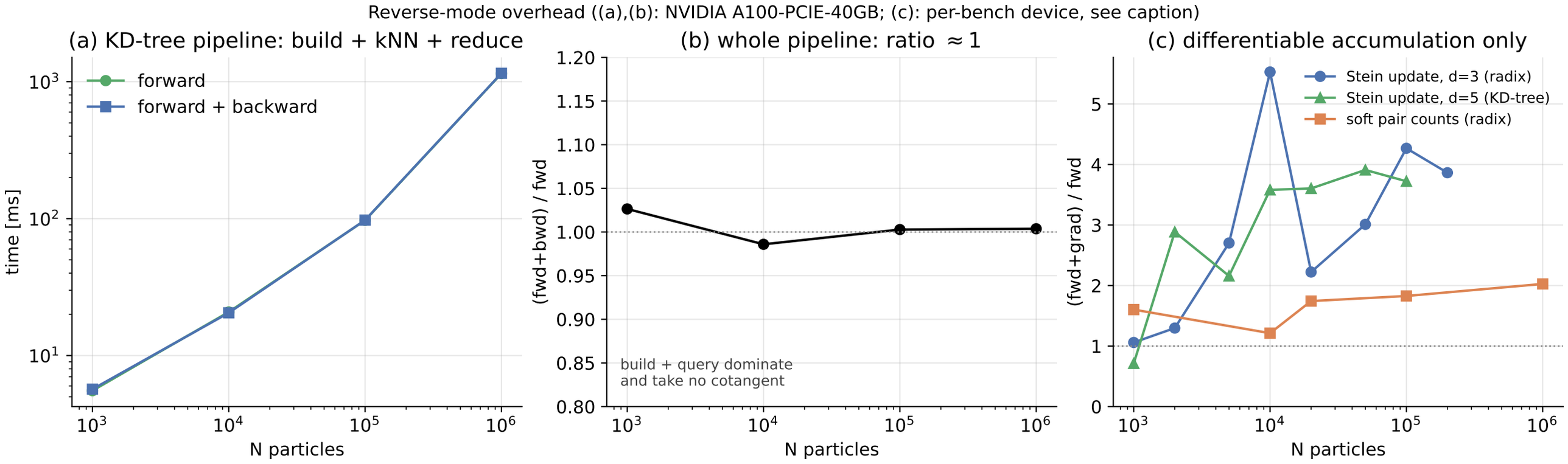
Reverse-mode overhead ((a),(b): NVIDIA A100-PCIE-40GB; (c): per-bench device, see caption)
(a) KD-tree pipeline: build + kNN + reduce
forward
forward + backward
time [ms]
N particles
(b) whole pipeline: ratio ≈ 1
(fwd+bwd) / fwd
build + query dominate
and take no cotangent
N particles
(c) differentiable accumulation only
Stein update, d=3 (radix)
Stein update, d=5 (KD-tree)
soft pair counts (radix)
(fwd+grad) / fwd
N particles